Is the VLA Backbone Enough to Read Human Goals? Early Goal Disambiguation from Egocentric Video

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***Abstract*—Collaborative robots must often infer what a person is trying to achieve before the action is finished. Vision–language–action (VLA) policies such as SmolVLA receive the goal only through a language instruction and have no component that estimates the human partner's intention. We study an intention module that would feed such a policy, and ask a concrete design question: should the module reuse the VLA's own visual backbone, or rely on an image–text encoder that is already aligned with language? We evaluate both on *early goal disambiguation*: from only the first $k \in \{2, 4, 6, 8\}$ frames of an egocentric manipulation step, rank the true purpose sentence above other purposes from the same video. On 2,503 steps from 27 videos and 13 scenes of EgoIntent, with 8-fold scene-grouped cross-validation, video-level bootstrap intervals and pre-registered decision rules, a readout trained on frozen SmolVLM2 features reaches only 1.13–1.19$\times$ chance MRR (intervals include 1) and is at chance on a locked test split. A zero-shot SigLIP encoder with no trained parameters reaches 2.25–2.57$\times$ chance MRR (all lower bounds ≥ 1.87) and improves with more observed frames. Nested selection over 90 training-free configurations picks the same pipeline in 8/8 folds (SigLIP so400m, mean pooling, subtraction of earlier-step video context), reaching 34.1–38.7% 5-way accuracy (20% chance). Frozen and evaluated once on a locked test split, it stays above chance at 4/4 prefixes and above the base SigLIP pipeline at 3/4, but drops to 21.9–30.2%. A top-2 margin lets the module abstain: answering the 50% most confident queries adds 6.9–7.9 points. Finally, SmolVLA on LIBERO-10 succeeds in 19/100 episodes with the correct instruction and 0/100 with an empty or wrong one, which motivates an intention channel; we provide a tested interface that injects the intention vector into SmolVLA as one prefix token.**

***Index Terms*—intention prediction, egocentric video, vision–language–action models, human–robot collaboration, zero-shot retrieval**

I. Introduction

A robot that helps a person at a table is useful only if it acts on the person's goal, and is most useful if it can do so *before* the goal becomes obvious. The difficulty is early ambiguity: lifting a cup is compatible with storing it, filling it, or putting it in a tray to be washed, and the observed motion diverges only later.

Vision–language–action (VLA) models [1]–[4] map camera images, a language instruction and the robot state to motor commands. They inherit broad visual and linguistic knowledge from a pretrained vision–language backbone, but the goal reaches them only through the instruction. In collaboration, the human rarely states every intention, so the robot needs an *intention module* that turns its observation of the human into a goal signal the policy can consume.

This paper studies how to build such a module. A natural choice is to reuse the VLA's own backbone: if the policy already understands images, a light readout on its features might recover human goals. An alternative is a contrastive image–text encoder [14], [15], which is trained so that images lie close to matching sentences. We compare the two under a protocol designed to be hard to fool, and make three contributions:

- An evaluation protocol for *early goal disambiguation* from egocentric video: strictly causal prefixes, scene-grouped cross-validation, same-video distractors, exact chance levels, video-level bootstrap, and decision rules fixed before running.
- Evidence that a readout trained on frozen SmolVLM2 features is close to chance on unseen scenes, whereas zero-shot SigLIP gives roughly twice the lift; a nested-selected training-free pipeline with causal context subtraction reaches 34–39% 5-way accuracy and keeps its advantage on a one-shot locked test.
- VLIA, a minimal interface that injects an intention vector into SmolVLA as one prefix token, with implementation tests for three intention sizes, and a LIBERO ablation showing that SmolVLA fails completely without goal information. Behavioural benefit of a predicted intention is not claimed.

II. Related Work

Egocentric anticipation and intention. Large egocentric corpora [8], [9] enabled long-term action anticipation. Intention has been modelled as a latent variable conditioning future actions [10], inferred with large language models [11], or used to guide reasoning [12]. These works predict sequences of future action labels. We instead isolate a narrower question

that matters for assistance: discriminating among *competing purposes* inside the same activity from a short prefix. EgoIntent [13] provides the required pre-outcome WHAT/WHY annotations.

VLA policies. RT-1/RT-2 [1], [2] and Open X-Embodiment [3] showed that transformer policies on vision–language features generalise across tasks. SmolVLA [4], built on LeRobot [5], couples a 500M SmolVLM2 backbone [6] with a flow-matching [7] action expert. None of these models estimate the intention of a human partner.

Image–text alignment. CLIP [14] and SigLIP [15] learn a joint embedding from web-scale image–caption pairs and transfer zero-shot to new label sets. Our results suggest this alignment, rather than the visual features alone, is what a small-data intention module lacks.

III. Problem and Method

A. Early goal disambiguation

Each manipulation step i has a video segment ending before its outcome, a WHAT label (the visible sub-action) and a WHY label y_i (a short free-form purpose sentence). Eight frames $x_{i,1:8}$ are sampled uniformly; a prefix of length k exposes only $x_{i,1:k}$. An intention module consists of an observation encoder f and a sentence encoder g and scores candidates by

$$s(i, c) = \cos\big(f(x_{i,1:k}),\, g(c)\big). \tag{1}$$

We use two candidate sets: (a) *gallery ranking* over all WHY sentences of the held-out fold (several hundred), and (b) K-*way choice* between y_i and $K-1$ different WHY sentences of the *same video*. The latter is the assistance-relevant case: distractors share scene, objects and often the opening motion.

B. Intention encoders

Trained readout on the VLA backbone. Each frame is encoded by frozen SmolVLM2-500M-Video-Instruct (960-d). The first k frame vectors are mean-pooled and passed through a two-layer MLP (512 hidden units). Targets are either a frozen sentence encoder (all-MiniLM-L6-v2 [16], `sentence`) or SmolVLM2 text features reduced to 256-d by PCA (`smolvlm_pca`). Training uses multi-positive InfoNCE ($\tau = 0.07$) plus $0.1\times$ a cosine loss, AdamW (lr 10^{-3}, weight decay 10^{-4}), up to 100 epochs, three seeds. Mean pooling is used because GRU and Transformer temporal heads did not outperform it in preliminary runs.

Zero-shot SigLIP. We encode frames with the image tower of `siglip-base-patch16-224` [15], ℓ_2-normalise, mean-pool the first k frames and renormalise; candidate sentences are encoded with the text tower (max 64 tokens; none truncated). No parameter is trained on our data. Note that SmolVLM2 itself uses a SigLIP-family vision tower, but its weights are adapted for generation and it carries no aligned text tower; the checkpoint we use is a different one.

Nested-selected pipeline. The zero-shot encoder leaves training-free choices open. We search a grid of 5 encoders (SigLIP base 224/384, SigLIP2 base, SigLIP so400m, CLIP ViT-L/14) $\times$ 3 frame aggregators (mean; max of per-frame cosine; log-sum-exp) $\times$ 3 centerings $\times$ 2 text templates (raw WHY; "a first-person photo of a person doing: {why}"). Centering C1 subtracts training-fold mean image and text embeddings (a modality-gap correction). Centering C2 subtracts the mean embedding of all frames of *strictly earlier* steps of the same video, falling back to C1 when there is none (1% of queries). C2 removes what the scene and activity share across steps and keeps what is new in the current one, while remaining causal. In each outer fold, the configuration with the best mean 5-way accuracy on the *training* folds is applied to the held-out fold.

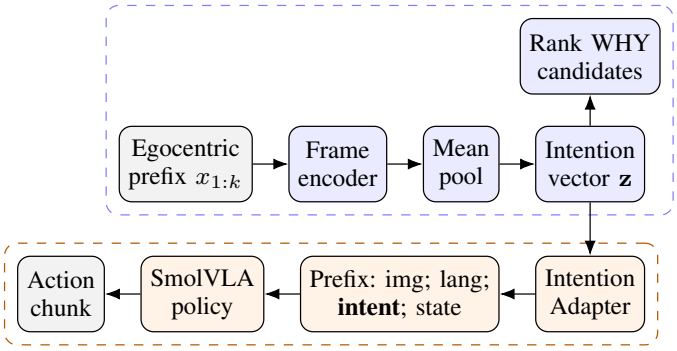


Fig. 1. VLIA. The intention module (top, evaluated here) maps an egocentric prefix to an intention vector; the adapter turns it into one prefix token of SmolVLA (bottom, implementation-tested only).

Selective prediction. The module may abstain when the top-1 minus top-2 cosine margin among the $K = 5$ candidates is below a threshold; thresholds for each fold are set from the other folds only.

C. Injecting intention into SmolVLA

In the VLIA architecture (Fig. 1), the intention vector $\mathbf{z} \in \mathbb{R}^d$ ($d = 256$, 768 or 1152) is projected by a linear Intention Adapter to the 960-d hidden space of SmolVLA and inserted as one token before the state token in the multimodal prefix, $[E_{\text{img}}; E_{\text{lang}}; W\mathbf{z}{+}b; E_{\text{state}}]$. Because the prefix is cached before the flow-matching steps, the extra token adds negligible cost, and disabling it recovers the original policy exactly.

IV. Experimental Protocol

Data. We use the train and validation portions of EgoIntent [13], derived from Ego4D: 2,503 steps, 27 source videos, 13 scenes (e.g. kitchen, workshop, farm, laundry room), 2,032 distinct WHY sentences, 19–125 steps per video.

Scene-grouped cross-validation. Scenes are assigned to 8 folds of similar size; every video belongs to one fold, and each step is evaluated once, out-of-fold, in a scene the model has not seen. For the trained readout, the number of epochs is chosen on an inner 15% split of the *training* scenes, after which the model is refit on all training scenes.

Metrics and chance. We report MRR and Top-n on the gallery with exact-text positives, and K-way accuracy for $K \in \{5, 10\}$ with distractors fixed per query by a hash of its id

(identical across methods, prefixes and seeds; all 2,503 queries valid). Gallery chance is computed exactly per query from fold size and number of positives (mean MRR chance 0.0298). *Lift* is the ratio of a metric to its chance level.

Uncertainty. 95% intervals come from 10,000 bootstrap resamples of whole source videos. Prefix comparisons use paired per-example differences.

Pre-registered rules. Written to the repository before the corresponding runs: (R1) the readout is above chance if the lower bound of its lift exceeds 1; (G1) SigLIP passes if, at every k, the lower bounds of MRR lift and 5-way lift exceed 1 *and* 5-way accuracy is at least 0.30; (Gate) an HOI-crop variant is evaluated only if MediaPipe [17] detects hands in $\geq$60% of frames overall and $\geq$30% in every scene; (N1) nested selection helps if its paired gain over base SigLIP has lower bound > 0 for both MRR and 5-way at $\geq$2/4 prefixes; (N2) it meets the practical bar if 5-way accuracy is $\geq$0.30 and MRR and 5-way lift lower bounds exceed 1 at $\geq$2/4 prefixes; (T1) on the locked test the frozen pipeline's 5-way lift lower bound exceeds 1 at $\geq$3/4 prefixes; (T2) its paired 5-way gain over base SigLIP has lower bound > 0 at $\geq$2/4; (S1b) accuracy at 50% coverage minus full coverage has lower bound > 0 at $\geq$2/4; (L1/L2) on LIBERO, success(correct)$-$success(empty) and success(empty)$-$success(wrong) have lower bound > 0.

Locked test. The readout was additionally evaluated once on the EgoIntent test portion (407 steps, 4 videos, 2 scenes) under a protocol locked in advance. One test video that overlapped an earlier pilot validation set was removed before locking. This split is video- but not scene-disjoint. The nested-selected pipeline was then frozen into a hashed configuration file, together with rules T1/T2, and run once on the same 407 steps; context subtraction on test uses only earlier steps of the same test video. With 4 videos, test intervals rest on 4 bootstrap clusters and are exploratory.

LIBERO ablation. A SmolVLA checkpoint retrained on LIBERO-10 [18] (7-D actions, 25k steps) is run on 10 tasks $\times$ 10 episodes under three instructions: correct, empty, and the instruction of task $(t + 5) \bmod 10$. Episode e uses the same initial state and seed in every condition.

V. Results

A. The trained readout is close to chance

Table I shows that the readout on frozen SmolVLM2 features does not satisfy R1 for MRR at any prefix, for either target: every lower bound is below 1. On 5-way choice, `smolvlm_pca` has lower bounds above 1 at all prefixes, but accuracy is only 21.9–23.0% against 20% chance. The two targets are statistically indistinguishable (paired lower bound above 0 at 0/4 prefixes). Heterogeneity is broad: depending on k, 8–12 of 27 videos fall below chance for `sentence`. On the locked test split, the best mean MRR (0.0314 at $k = 8$) is at the gallery chance of 0.0318, although validation MRR on the fixed split had been 1.7–1.9$\times$ chance with epochs selected on that same split. This gap is what motivated scene-grouped cross-validation.

TABLE I
SCENE-HELD-OUT RESULTS (8 FOLDS, 2,503 QUERIES). READOUT VALUES ARE MEANS OVER 3 SEEDS; BRACKETS ARE 95% VIDEO-BOOTSTRAP INTERVALS. CHANCE: MRR 0.0298, 5-WAY 0.20.

Encoder	k	MRR	MRR lift	5-way	5-way lift
Readout (`sentence`)	2	.0335	1.13 [0.86, 1.43]	.216	1.08 [1.00, 1.16]
	4	.0345	1.16 [0.88, 1.48]	.215	1.07 [1.00, 1.16]
	6	.0353	1.18 [0.90, 1.52]	.212	1.06 [0.98, 1.16]
	8	.0347	1.17 [0.87, 1.51]	.219	1.10 [1.01, 1.20]
Readout (`smolvlm_pca`)	2	.0340	1.14 [0.95, 1.36]	.219	1.09 [1.02, 1.17]
	4	.0338	1.14 [0.94, 1.35]	.222	1.11 [1.03, 1.19]
	6	.0346	1.16 [0.95, 1.39]	.223	1.12 [1.04, 1.20]
	8	.0354	1.19 [0.98, 1.42]	.230	1.15 [1.07, 1.23]
SigLIP zero-shot	2	.0671	2.25 [1.87, 2.75]	.277	1.39 [1.23, 1.56]
	4	.0715	2.40 [2.02, 2.88]	.275	1.38 [1.22, 1.55]
	6	.0734	2.47 [2.06, 2.97]	.281	1.40 [1.24, 1.58]
	8	.0765	2.57 [2.13, 3.14]	.289	1.44 [1.27, 1.62]

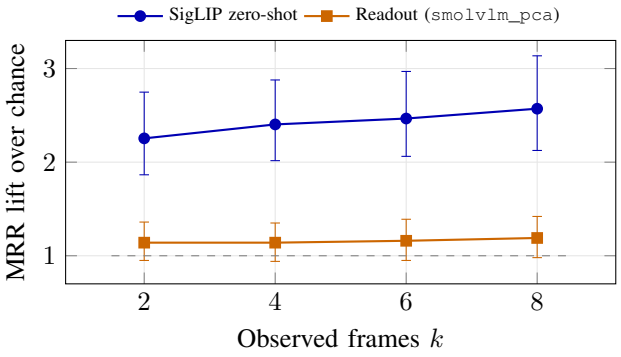


Fig. 2. MRR lift versus prefix length with 95% video-bootstrap intervals. Dashed line: chance.

B. Zero-shot alignment roughly doubles the lift

With no trained parameters, SigLIP reaches 2.25–2.57$\times$ chance MRR, and its interval lower bounds (1.87–2.13) lie above the readout's upper bounds (1.35–1.52) at every prefix (Fig. 2). 5-way accuracy rises to 27.5–28.9% and 10-way to 15.7–17.7% (chance 10%). Under G1, however, SigLIP **fails**: both interval conditions hold at 4/4 prefixes, but 5-way accuracy ≥ 0.30 holds at 0/4. We keep this verdict.

C. More observation helps only with aligned features

Paired over identical queries, SigLIP's MRR increases by $+0.0094$ [$+0.0053$, $+0.0141$] from 2 to 8 frames (about 14% relative) and 5-way accuracy by $+0.0112$ [$+0.0014$, $+0.0211$]. For the readout, all paired MRR differences have intervals containing 0 (e.g. $+0.0012$ [-0.0014, $+0.0039$] for `sentence`, 2$\rightarrow$8). Additional visual evidence thus translates into better disambiguation only when the representation is already aligned with language.

D. Training-free selection and the locked test

In all 8 outer folds, nested selection chose the same configuration: SigLIP so400m, mean pooling, C2, template T1 (no ties). Table II compares it with base SigLIP. 5-way accuracy rises to 34.1–38.7% with paired gains of $+0.064$ to $+0.099$

TABLE II
5-WAY ACCURACY OF BASE SIGLIP (B) AND THE NESTED-SELECTED/FROZEN PIPELINE (F). CV: 8-FOLD OUT-OF-FOLD, 2,503 QUERIES. TEST: LOCKED SPLIT, 407 QUERIES, 4 VIDEOS, RUN ONCE (INTERVALS EXPLORATORY). Δ: PAIRED F$-$B WITH 95% VIDEO-BOOTSTRAP INTERVAL.

	Cross-validation			Locked test		
k	B	F	Δ	B	F	Δ
2	.277	.341	+.064 [.039, .089]	.204	.219	+.015 [−.010, .038]
4	.275	.349	+.074 [.049, .100]	.211	.246	+.034 [.013, .048]
6	.281	.362	+.082 [.051, .112]	.224	.278	+.054 [.010, .097]
8	.289	.387	+.099 [.062, .134]	.214	.302	+.089 [.044, .148]

(all lower bounds > 0), so **N2 passes**; MRR lifts are 2.51–2.86. The MRR gain, however, has intervals containing 0 at every prefix, so **N1 fails**. Replacing frames by those of another scene's video ($k = 8$, 313 queries) drops accuracy from 39.6% to 18.8%. In the descriptive grid, the top six configurations all combine so400m with C2; removing C2 gives 31.1–33.0%.

On the locked test, the frozen pipeline's 5-way lift lower bound exceeds 1 at 4/4 prefixes (**T1 passes**) and its paired gain over base SigLIP is positive at 3/4 (**T2 passes**); 10-way gains are positive at 4/4. Absolute accuracy is lower than in cross-validation (21.9–30.2%), and base SigLIP drops as well (20.4–22.4%). Gallery MRR of the frozen pipeline is *below* base SigLIP at every prefix (interval below 0 at $k = 2$): context subtraction helps separate steps within a video, which is the assistance-relevant case, but not ranking across videos.

E. Knowing when to wait

With the margin of the selected pipeline as confidence score, accuracy at 50% coverage exceeds full-coverage accuracy by $+0.069$ to $+0.079$ (lower bounds ≥ 0.039; **S1b passes**), and at 20% coverage it reaches 53–61%. Leakage-free fold thresholds realise the target coverages (0.50, 0.20) with 40.7–46.1% and 53.3–60.9% accuracy. The same analysis on base SigLIP gains only 2.6–3.5 points. Two other pre-registered ideas failed: stopping observation at the first frame whose margin exceeds a threshold (S2) did not beat fixed-k at matched frame budgets, and a kNN text-space correction from training scenes (K1) lowered MRR.

F. Does SmolVLA need the goal?

On LIBERO-10, SmolVLA succeeds in 19/100 episodes with the correct instruction (Wilson [0.125, 0.278]) and in 0/100 with an empty or a wrong one (both [0, 0.037]); all failed episodes run to the 520-step limit. **L1 passes** ($+0.19$ [$+0.12$, $+0.27$]); **L2 fails**, since both alternatives are at zero. Empty and wrong prompts are out of distribution, so this measures dependence on the language channel, not the value of a predicted intention.

G. Diagnostics

Paraphrase ceiling. WHY sentences are free-form: 66.6% of queries have another WHY in the same video with sentence-embedding cosine ≥ 0.8 (44.5% at ≥ 0.9), yet only exact text counts as correct. Two pre-registered secondary variants estimate the effect: with distractors filtered to cosine < 0.8, SigLIP's 5-way accuracy at $k = 8$ is 30.8%; counting near-paraphrases (≥ 0.9) as positives gives MRR 0.112 (lift 2.13 [1.71, 2.66]). These do not change the G1 verdict.

Typical failure. For a dough-kneading step labelled "positioning basin", SigLIP ranks "prepare to knead dough", "developing dough consistency" and "forming cohesive dough" above it: the activity is recognised, the step-level purpose is not. This is the characteristic error of the task, since same-video candidates share objects and scene.

HOI gate. MediaPipe detects hands in 58.2% of 20,024 frames, with four scenes below 30% (deck 10.3%, laundry room 15.3%, yard 15.9%, outdoor space 24.8%). The gate fails, so no HOI-crop variant was evaluated and we draw no conclusion about it.

H. Policy interface

The Intention Adapter and prefix injection pass implementation tests with 256-, 768- and 1152-d intention vectors: gradients reach the adapter, the prefix grows by exactly one token placed before the state token, the token is present in the KV cache at inference, the flow-matching loss is finite, checkpoints round-trip, and disabling intention reproduces the original SmolVLA. We do not claim that intention improves robot behaviour.

VI. DISCUSSION AND LIMITATIONS

The main lesson is a design one: with about two thousand labelled steps, an intention module cannot learn the visual–linguistic mapping from the VLA backbone's visual features, while a pretrained aligned encoder provides it for free and is the better default for the intention branch. Scoring WHY sentences with SmolVLM2's own likelihood (PMI against a no-image prompt) confirms this: it is above chance (5-way 23.5–24.6%) but 4.2 points below base SigLIP on identical queries. Subtracting causal video context is a cheap, training-free addition that helps exactly in the within-activity setting a collaborative robot faces. Limitations are substantial. The main study covers 27 videos in 13 scenes; the locked test has only 4 videos in 2 scenes, is not scene-disjoint, and shows a clear drop from cross-validation. The selected pipeline lowers cross-video gallery MRR on test. Selective prediction has not been tested on the locked split. The LIBERO ablation uses one checkpoint and 10 episodes per task. Closed-loop evaluation with a predicted intention token is future work.

VII. CONCLUSION

We evaluated intention modules for VLA policies on early goal disambiguation from egocentric video under scene-held-out, pre-registered conditions. A readout on the VLA's frozen backbone features is near chance, whereas zero-shot image–text alignment yields roughly twice the lift. A training-free pipeline with causal context subtraction, chosen by nested selection, reaches 34–39% 5-way accuracy and keeps a smaller but positive advantage on a one-shot locked test, and its

confidence margin supports abstention. Since SmolVLA fails without goal information, the next step is to train it with intention tokens on LIBERO and measure how much of the lost success they recover, then move to scene-disjoint test data and a UR5 arm. Code and result files are available at https://github.com/sgdfgnhmjnfd/vlia-uet.

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